W10 · Original Q3–Q5 Question Workbook

Advanced Machine Learning · November3,2026. Original wording, assumptions and original points are retained. The points are archival labels; this is formative class practice, with no new graded assignment. Write your own derivations in the blank grids.

Q3 · Importance Sampling and Effective Sample Size

20 points

For $f(z) = z$, the normalized target distribution p and proposal distribution q are:

z	0	1	2
$p(z)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
$q(z)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

Four independent draws from q are $(0, 1, 1, 2)$; count repeated values separately. Use

$$w_i = \frac{p(z^{(i)})}{q(z^{(i)})}, \quad W_i = \frac{w_i}{\sum_{j=1}^{4} w_j}, \quad N_{\text{eff}}^{\text{weight}} = \frac{1}{\sum_{i=1}^{4} W_i^2}.$$

Q3(a)

Compute the raw weights w_i, their sum, and normalized weights W_i, in draw order. **[8 points]**

Q3(a) · Your working and explanation

Q3(a) · Your working and explanation

Q3(b)

Compute the exact target expectation $\mathbb{E}_p[z]$, the direct IS estimate, and the self-normalized IS (SNIS) estimate from these four draws. Show the denominator of each estimate. **[8 points]**

Q3(b) · Your working and explanation

Q3(c)

Compute the weight ESS using the normalized weights from (a). Give both an exact fraction and a decimal value. **[4 points]**

Show all working and final answers for (a)-(c).

Q3(c) · Your working and explanation

Q3(c) · Your working and explanation

Q4 · Predictive Loss and Paired Comparisons

20 points

Three responses are conditionally independent, with $y = (0, 2, 4)$, predicted means $\mu_A = (1, 1, 3)$ and $\mu_B = (0, 1, 4)$. Both models use the same known variance $\sigma^2 = 2$:

$$p(y_i \mid \mu_i, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - \mu_i)^2}{2\sigma^2}\right].$$

For (c), use these separate experiment records. Match conditions by run ID, not row position. Lower error is better.

A: run ID	A: error	B: run ID	B: error
1	0.4	3	0.2
2	0.3	1	0.3
3	0.5	2	0.4

Q4(a)

Derive the summed Gaussian NLL in terms of $\mathrm{SSE} = \sum_{i=1}^{3} (y_i - \mu_i)^2$, retaining the normalizing constant. Use natural logarithms. **[6 points]**

Q4(a) · Your working and explanation

Q4(a) · Your working and explanation

Q4(b)

Calculate SSE and MSE for each model, and $\mathbf{NLL}_A - \mathbf{NLL}_B$. Show residual calculations. **[8 points]**

Q4(b) · Your working and explanation

Q4(c)

Match by run ID. Calculate $d_i = e_{A,i} - e_{B,i}$ for runs $1, 2, 3$, their mean, and the number of runs B wins. A tie is not a win. **[6 points]**

Show all working and final answers for (a)-(c).

Q4(c) · Your working and explanation

Q4(c) · Your working and explanation

Q5 · Bayesian Prediction in Weight and Function Space

20 points

Let $f(x) = xw$, with $w \sim \mathcal{N}(0, \frac{1}{2})$ and $y_i = x_i w + \epsilon_i$, where the training noises $\epsilon_i \sim \mathcal{N}(0, 1)$ are independent of each other and of w. The second argument of $\mathcal{N}$ is the **variance**. Observe $D = \{(1, 1), (2, 3)\}$.

Q5(a)

Find the posterior mean and variance of $w \mid D$. Show the prior–likelihood calculation, using either square completion or a precision update. **[8 points]**

Q5(a) · Your working and explanation

Q5(a) · Your working and explanation

Q5(b)

For a **new response** at $x_* = 2$, let $f_* = 2w$ and $y_* = f_* + \epsilon_*$, where $\epsilon_* \sim \mathcal{N}(0, 1)$ is independent of w and the training noises. Calculate $E[f_* \mid D]$, $\mathbf{Var}(f_* \mid D)$ and $\mathbf{Var}(y_* \mid D)$. Show the variance calculations. **[8 points]**

Q5(b) · Your working and explanation

Q5(c)

Before observing D, derive the prior kernel $k(x, x') = \mathrm{Cov}(f(x), f(x'))$ and calculate $k(-1, 2)$. Show how the weight prior determines this covariance. **[4 points]**

Show all working and final answers for (a)-(c).

Q5(c) · Your working and explanation

Q5(c) · Your working and explanation